

Yuelong ZHANG

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Research Interests

Bio-Inspired Robotics, Legged Locomotion, PEA Design, and Learning-Based Motion Control

Education

Zhejiang University, Haining, Zhejiang, China

Aug. 2024 – Jun. 2028 (expected)

B.Eng. in Mechanical Engineering, ZJU-UIUC Institute

- GPA: 3.84/4.00
- Class Ranking: 6/61

University of Illinois Urbana-Champaign, Urbana, IL, USA

Aug. 2024 – Jun. 2028 (expected)

B.S. in Mechanical Engineering

- GPA: 4.00/4.00

Research Experience

Intelligent Robot Vision & Control Lab, ZJU-UIUC Institute, Zhejiang University

Oct. 2025–Present

Undergraduate Research Assistant

Advisor: Prof. Liangjing YANG

- Research project: *Research on the Elastic Subsystem of Bionic Manipulator for Flexible Drive and Energy Recovery.*
- Robotic mechanisms design, RL Training, Robot Controlling for bio-inspired robotic systems.

State Key Laboratory of Fluid Power and Mechatronic Systems, Zhejiang University

Jun. 2025–Present

Undergraduate Research Assistant

Advisors: Prof. Tao LIU and Prof. Jingang YI

- Research project: *Bionic Interactive Reinforcement Learning and Natural Gait Transfer Control Strategies for Bipedal Robots.*
- Robot designs and RL pipelines for AMP optimization, locomotion control, and natural gait transfer.

Manuscripts

1. Wenyu Li, Jiaying Li, **Yuelong Zhang**, Zhiyuan Xu, Jingang Yi*, and Tao Liu*, “**BioLeg: A Mammalian Locomotion-Inspired Bipedal Robot.**”

Submitted to *IEEE/ASME Transactions on Mechatronics*.

Status: Under review. *Corresponding authors.

Technical Skills

Engineering Skills: CAD Modeling, Reinforcement Learning, AMP Optimization

Software & Platforms: MATLAB, Python, Simulink, SolidWorks, Fusion 360, MuJoCo, Isaac Lab, ROS,

Languages: Chinese (Native), English (Proficient)

References

Prof. Liangjing YANG

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Prof. Tao LIU

Professor, School of Mechanical Engineering, Zhejiang University

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Prof. Jingang YI

Professor, School of Aerospace & Engineering, Rutgers University

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